Soft fragment–loss balancing closes the seed-42 neural MS2LDA motif gap

MS2LDA research checkpoint

22 August 2026

Abstract

We tested one small correction to the document-gated neural MS2LDA architecture after diagnosing that its topics were strongly divided by token type: 453 of 1,000 topics had no fragment among their 20 highest-probability words. The candidate preserves word rankings within fragments and neutral losses while pulling each type's total topic mass 25 percent toward an equal split. It uses the same one-pass router and a fixed three-quarter document gate. Seed 42, 1,000 topics, six CPU threads, data, token features, initialization, losses, and final epoch 40 remain fixed. On leakage-filtered validation data, useful high-confidence motifs—topics associated at probability at least 0.5 with compound-balanced structure overlap score (SOS) at least 0.6—increased from 20 to 148, compared with 138 for Tomotopy. MAG annotation coverage reached 51.5 percent, mean SOS was 0.6418, and validation NLL was 8.4963. Every predeclared validation gate passed. The locked checkpoint was then evaluated once on test, where it produced 211 useful motifs versus Tomotopy's 186, with nearly identical mean SOS. This is a substantial single-seed research result, not multi-seed confirmation or evidence to replace the production backend.

1 Question and frozen comparison

MS2LDA treats spectra as documents, fragments and neutral losses as words, and recurring word distributions as Mass2Motifs [1]. The previous neural checkpoint had better completion likelihood and faster direct inference than the established Tomotopy model, but far fewer chemically annotated and useful high-confidence motifs. Adding document evidence raised confidence, yet the first gated candidate still optimized only 420 motifs and produced 83 useful high-confidence motifs, versus Tomotopy's 607 and 138.

The comparison fixes seed 42, 1,000 requested topics, six CPU threads, the training/validation split, vocabulary, token features, deterministic initialization, optimization schedule, losses, one-pass inference, and final epoch 40. During development the candidate run exposed named training and validation files only. Its MAG index excluded exactly the union of validation and test compound connectivity keys.

2 Failure mode and minimal correction

The failed gated model was not primarily short of confidence. Its topic-word distributions separated fragments from neutral losses too strongly. Among each topic's 20 highest-probability words, 453 topics contained no fragment and 330 contained only fragments; Tomotopy had just 27 fragment-free topics. Because MAG motif optimization benefits from joint fragment–loss evidence, increasing the gate weight alone raised useful motifs but did not repair annotation breadth.

Let m_k be the original total fragment probability in topic k. The candidate replaces it with

$$\widetilde{m}_k = (1-\lambda)m_k + \lambda/2, \qquad \lambda = 0.25.$$

It rescales fragment and loss probabilities to these two totals while preserving all rankings within each token type. This is a soft constraint: topics may remain fragment- or loss-heavy, but neither type is allowed to vanish as easily. It adds no learned parameter or loss.

After count-weighted token aggregation, document evidence is applied as

$$g_d = \mathrm{softmax}(\text{document logits}/T), \qquad \theta_d = \mathrm{normalize}(\text{token mass} \times g_d^{0.75}).$$

The forward calculation is identical in training and inference. During training, g_d is treated as fixed evidence so the shared router cannot improve the loss merely by strengthening its own gate. Multiplication preserves zero token support, and empty spectra retain the uniform fallback. Setting the type balance and gate weights to zero exactly recovers the earlier decoder and token mixture.

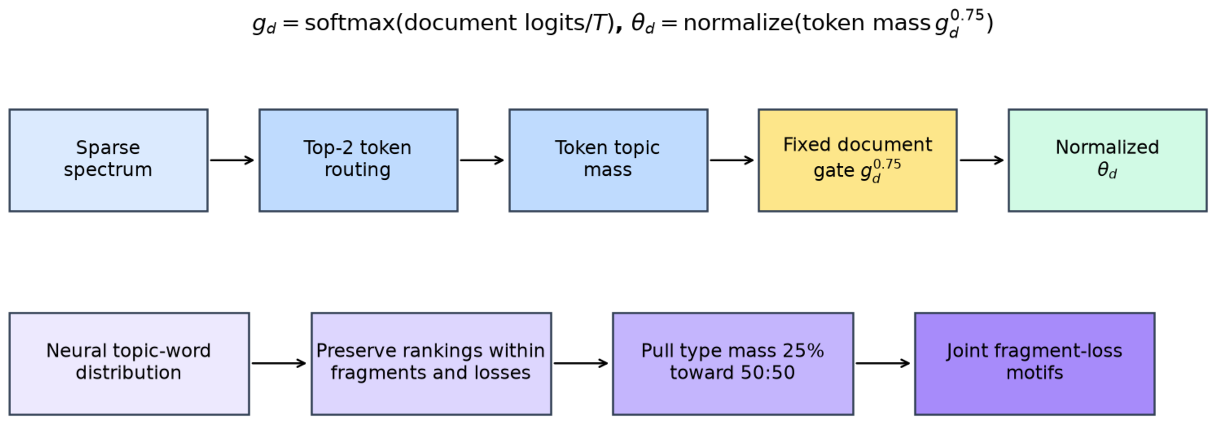

Figure 1: The accepted correction has two parameter-free parts: a fixed document gate for confident spectrum mixtures and a soft token-type balance for joint fragment–loss motif content.

3 Paper-aligned validation

MAG annotation and SOS use validation spectra only. Library entries sharing a validation or test connectivity key are excluded before retrieval. Topic association follows the paper threshold, $\theta_{dk} \geq 0.5$. SOS is averaged over unique compounds within each topic. Eligible topics are divided into high (> 0.8), intermediate (0.6–0.8, inclusive), and low (< 0.6) bands.

The primary outcome is the number of high-confidence motifs with SOS at least 0.6. Acceptance required all of the following:

- closing at least 50 percent of the current-neural–Tomotopy useful-motif gap;
- MAG annotation coverage of at least 49.6 percent;
- mean high-confidence SOS no more than 0.02 below the current neural model;
- validation NLL no greater than 8.5266; and
- finite, stable training and evaluation.

The initially proposed 5,571-second fitting-time condition had already been demoted to descriptive context. The candidate nevertheless completed in 3,571.5 seconds and would pass that original condition.

4 Validation results

Table 1: Validation-only paper-relevant comparison.

Metric	Current neural	Balanced-gated neural	Tomotopy
Optimized motifs	384	515	607
High-confidence SOS-evaluable motifs	42	252	206
SOS high (> 0.8)	10	43	58
SOS intermediate (0.6–0.8)	10	105	80
SOS low (< 0.6)	22	104	68
Useful high-confidence motifs (SOS ≥ 0.6)	20	148	138
Mean high-confidence SOS	0.6227	0.6418	0.6761
Median high-confidence SOS	0.5866	0.6477	0.6854
Model-fitting time (minutes)	84.4	59.5	76.0

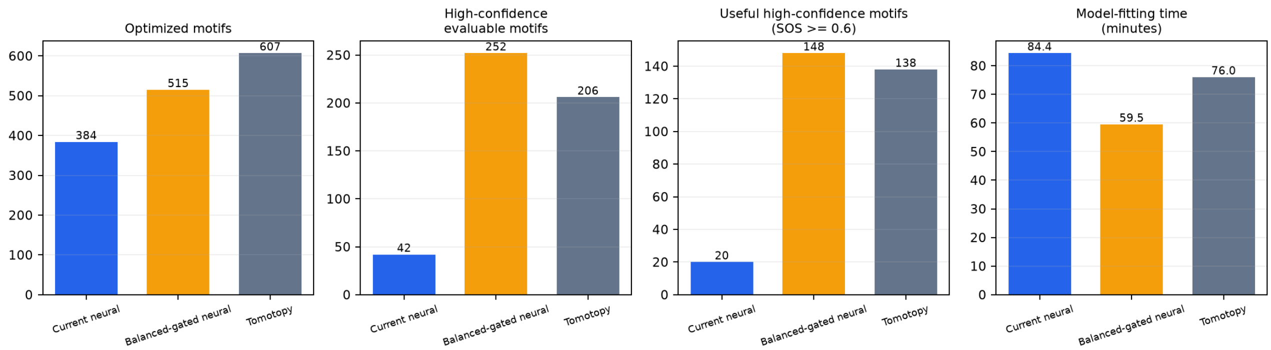

Figure 2: Headline validation outcomes and model-fitting time. The accepted model exceeds Tomotopy's useful high-confidence motif count while retaining the lower annotation coverage of the neural family.

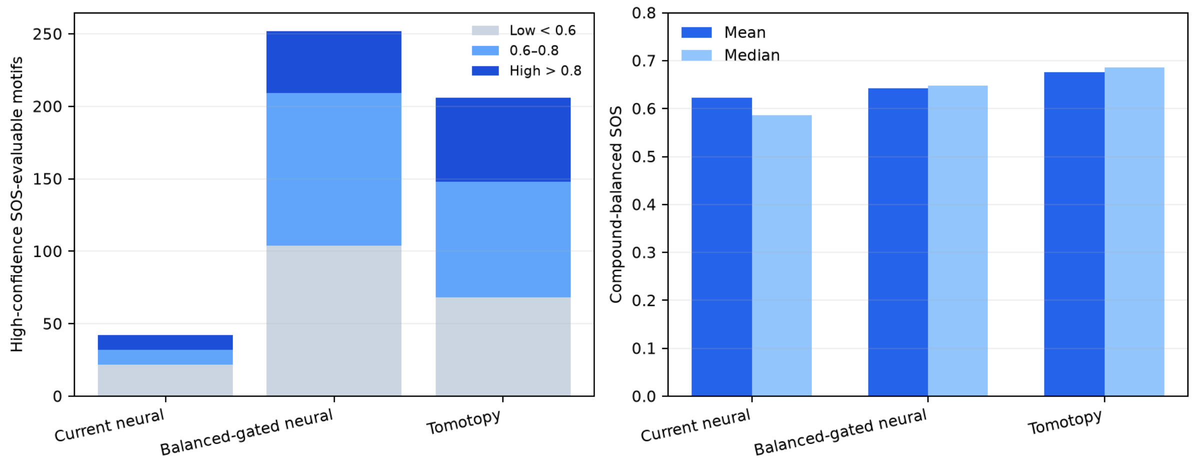

Figure 3: Validation high-confidence chemical evidence. Left: SOS bands among evaluable motifs. Right: compound-balanced mean and median SOS.

The candidate increases high-confidence associated validation spectra from 87 to 1,011 and SOS-evaluable high-confidence motifs from 42 to 252. Tomotopy has 446 associated spectra and 206 evaluable motifs. Useful motifs increase from 20 to 148, ten more than Tomotopy's 138. Relative to the original 118-motif gap, the candidate closes 108.5 percent.

Candidate mean SOS is 0.6418 versus 0.6227 for the current neural model, and median SOS rises from 0.5866 to 0.6477. Tomotopy remains higher on validation mean and median SOS at 0.6761 and 0.6854. The candidate's 51.5 percent MAG annotation coverage clears the 49.6 percent gate but remains 9.2 percentage points below Tomotopy's 60.7 percent.

Validation NLL is 8.4963, 1.64 percent above the current neural 8.3594 but below the 8.5266 safety limit. Training remains finite and stable with 313 corpus-active topics. Fitting takes 59.5 minutes, versus 84.4 minutes for the current neural reference and 76.0 minutes for the reused Tomotopy source run.

5 Single locked test confirmation

All validation checks passed before the test matrices and records were linked into the candidate run. The epoch-40 checkpoint hash was fixed, after which one neural test inference and one leakage-filtered chemistry evaluation were performed. No parameter changed after test access.

Table 2: One-time held-out test confirmation.

Metric	Balanced-gated neural	Tomotopy
Optimized motifs	515	607
High-confidence SOS-evaluable motifs	345	333
SOS high (> 0.8)	58	66
SOS intermediate (0.6–0.8)	153	120
SOS low (< 0.6)	134	147
Useful high-confidence motifs (SOS ≥ 0.6)	211	186
Mean high-confidence SOS	0.6376	0.6369
Median high-confidence SOS	0.6406	0.6145
Held-out completion NLL	8.5114	9.7569

The test result confirms rather than reverses the validation conclusion. The candidate produces 211 useful high-confidence motifs versus Tomotopy's 186 and 345 SOS-evaluable motifs versus 333. Mean SOS is effectively equal (0.6376 versus 0.6369), while the candidate median is higher (0.6406 versus 0.6145). Candidate completion NLL is 8.5114 versus Tomotopy's 9.7569. Annotation coverage remains lower at 51.5 versus 60.7 percent.

6 Secondary inference context

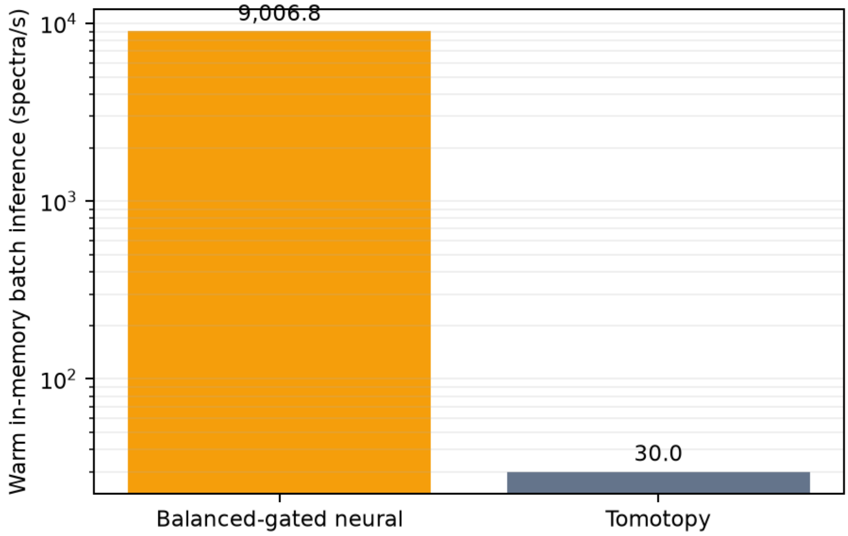

Figure 4: Matched six-thread warm in-memory batch inference on the held-out test matrices. This secondary result is not an acceptance gate or end-to-end application latency.

The accepted one-pass neural checkpoint processes 9,007 spectra/s in the fixed 128-spectrum warm batch, versus 30.0 spectra/s for Tomotopy with 100 inference iterations, a 300-fold ratio. "Warm in-memory batch inference" means that the model and batch are already loaded and one warm-up has completed. It is usable for serving or repeated batches, but excludes file loading, preprocessing, and first-call setup.

7 Interpretation

The main neural failure was not merely diffuse document mixtures. It was the combination of confidence with a polarized topic-word inventory. The fixed document gate supplies the former; soft fragment–loss balancing repairs the latter without a new encoder, loss, or iterative inference step. Their combination exceeds Tomotopy on the chosen useful-motif count on validation and test while maintaining predictive fit.

The remaining weakness is broad MAG annotatability: 515 of 1,000 neural motifs are optimized, compared with 607 for Tomotopy. Validation SOS among the useful set also remains somewhat lower. These are clear targets for later work, but they do not negate the present primary result. This remains one fixed-seed architecture experiment. Replication across seeds is a separate confirmation stage, and no claim is made that this research checkpoint replaces the production Tomotopy backend.

8 Reproducibility

The report retains the resolved comparison summary and hashed source manifests for training, all validation references, the gate decision, the test-access record, and both test evaluations. The candidate checkpoint is fixed at epoch 40 with SHA-256:

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90d7cb11f4f8717a8b028130b118e3bf
47ddd136eb576b3f8a388745305be33d
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The test-access manifest confirms that selection was final before test.

References

[1] J. J. J. van der Hooft et al. "Topic modeling for untargeted substructure exploration in metabolomics." *Proceedings of the National Academy of Sciences*, 2016. https://pmc.ncbi.nlm.nih.gov/articles/PMC5137707/

[2] M. Torres Ortega et al. "Large-scale discovery and annotation of substructure patterns in mass spectrometry profiles." *Nature Communications*, 2026. https://www.nature.com/articles/s41467-026-75038-0